

Original Research Article

Genotyping of multi drug resistant *Bacteroides fragilis* group of clinical isolates from mangalore, south IndiaZomuanpui Colney^a, Beena Antony^{a,*}, Shruthi Kanthaje^b^a Department of Microbiology, Father Muller Medical College, Kankanady, Mangalore, Karnataka, 575002, India^b Yenepoya Research Centre, Yenepoya (Deemed to be University), Derlakatte, Mangalore, India

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ABSTRACT

Introduction: *Bacteroides fragilis* group, the most encountered anaerobic bacterium is emerging with resistance to antibiotics. This study explores the antibiogram and occurrence of resistance genes in isolates of *B. fragilis* group from clinical samples.**Method:** In this study the antimicrobial susceptibility test was done using commercially available E strip test and the results were recorded according to CLSI guidelines. Genotypic investigations were performed by conventional PCR to detect the target resistant genes.**Results:** Ceftriaxone, cefoxitin, clindamycin and imipenem were found to be the most resistant antimicrobials in E test method. Metronidazole has shown resistance in 7 strains in vitro while resistance *nim* genes were detected in 12 strains from 62 randomly selected isolates. Other resistance genes (*cfiA*, *ermF* and *cepA*) were expressed at 58%, 62.9% and 48.3% respectively, among these strains.**Conclusion:** *B. fragilis* group harbouring the resistant genes may not be fully expressed phenotypically. Hence, detection of these genes by PCR might be necessary for a pertinent conclusion.

1. Introduction

Members of *B. fragilis* group are opportunistic pathogen and the most frequently isolated species with a high degree of virulence potential [1, 2]. They cause serious infections and has high tendency to develop resistance to antibiotics which has been reported worldwide. The major causes of the emerging resistance are as follows -relying on empirical treatment, lack of routine surveillance and monitoring of anaerobic susceptibility testing which are insufficient and overlooked [3].

B. fragilis group exhibit numerous antibiotic resistance mechanisms compared to other anaerobes as revealed in literature [1,4,5]. Hydrolysis of β -lactam agents is the most common resistance mechanism against *Bacteroides* spp because of an overall increase production of constitutive β -lactamase [5–7]. The first report of the penicillin-destroying activity of *B. fragilis* was by Garrod in 1955 [8]. Narrow spectrum penicillin and cephalosporins are intrinsically resistant which produces endogenous penicillinase/cephalosporinase belonging to Amber molecular Class A which is encoded by *cepA* gene. [6,9] Metallo β -lactamase produced by some *B. fragilis* strains, belongs to Amber molecular Class B which is encoded by *cfiA* gene, has become a serious threat in the recent curative treatment, as they are able to hydrolyse carbapenems [6,7]. It has been

stated that *cfiA* gene are generally silent and the spontaneous arrangement of two novel insertion sequence element upstream IS1186 and IS1188 determined the expression and resistance of carbapenems [10–12]. *B. fragilis* is the only *Bacteroides* spp. In which metallo- β -lactamases have so far been reported [6].

Clindamycin was one of the first agent used for treatment of *B. fragilis* infections and has emerged as the most rapidly increased resistance due to overall consumption [13]. The resistance mechanism has been ascribed as posttranscriptional modification of the 23S rRNA by an adenine-specific N-methyltransferase (methylase) specified by *erm* (erythromycin ribosome methylation) gene [14]. Three distinctively related genes, *ermF*, *ermFU*, *ermFS*, is particularly responsible for resistance of macrolides-linocosamide-streptogramin B in *B. fragilis*. [5,13].

Metronidazole is the drug of choice for the treatment of anaerobic infection. The mechanism that results in reduced susceptibility to metronidazole is associated with the presence of a nitroimidazole reductase encoded by a *nim* (A-L) gene [15], which converts 4- or 5-nitroimidazole into 4- or 5-aminoimidazole, thus avoiding the formation of toxic nitroso radicals that are essential for antimicrobial activity [12,16, 17]. Incidence of metronidazole-resistant *B. fragilis* has been reported worldwide between 0.5% and 7.8% [17].

* Corresponding author.

E-mail address: beenafmmc@fathermuller.in (B. Antony).<https://doi.org/10.1016/j.ijmmb.2020.10.013>

The objective of this study was to determine the antibiogram and the occurrence of resistance genes such as *nim*, *cepA*, *ermF* and *cfiA* in *B. fragilis* group from various clinical isolates, encountered in Coastal Karnataka region. To the best of our knowledge, this report appears to be the first one of such kind, investigating 4 MDR genes in *B. fragilis* group by molecular analysis in India.

2. Subjects and methods

2.1. Material and method

2.1.1. Study design

Various clinical samples such as intra-abdominal infection, deep seated abscesses, female genitourinary tract which were suggestive of *B. fragilis* group infections were included in this prospective, analytical study. The study was carried out at a tertiary care hospital in Karnataka from January 2017–December 2019. It was also approved by Institutional Ethics Committee (FMMC/FMIEC/4229/2017).

2.1.2. Microbiological analysis

The various clinical samples were inoculated directly and also from supplemental RCM (Robertson's cooked medium) and sub cultured using *Bacteroides* Bile Esculin Agar, Neomycin Blood Agar and Brucella Blood Agar (Himedia Pvt Ltd, India). Metronidazole discs 5 µg were applied in each plate for presumptive identification of anaerobes. Special potency discs were applied (Vancomycin (5 µg), Kanamycin (1000 µg) and Colistin (10 µg)) to serve as an aid in the grouping i.e. *B. fragilis* resistant to all the 3 discs. The inoculated plates were incubated at 37 °C for 48 h in anaerobic condition using commercially available BD GasPak™ Gas Generating Container/Sachet. The strains were isolated and identified according to the standard procedures [18].

Isolates were subjected to antibiogram by determining Minimum Inhibitory Concentration (MIC) using gradient diffusion method of E-test (Ezy MIC™ Strips HiMedia Laboratories) on Brucella Blood Agar in anaerobic condition. MIC breakpoints were as per the interpretative criteria provided by CLSI guidelines as susceptible, intermittent and resistant [19].

2.1.3. Detection of antimicrobial resistance by conventional polymerase chain reaction

Pure isolates were suspended in 200 µl Phosphate buffer saline (PBS) buffer in a 1.5 µl eppendroff vials and genomic DNA was extracted using QIAamp DNA Mini kit (Qiagen, Hilden Germany). Detection of *nim*, *cepA*, *cfiA* and *ermF* genes was performed by conventional PCR using primer set given in Table 1. The reaction was assembled into 10 µl volume containing 5–10 nM of each primer (Bioserve, Hyderabad), 2X PCR Taq MasterMix/with dye (ABM, Life Expression India), ~200 ng/µl DNA template and nuclease-free H₂O up to 10 µl in a thin walled 0.2 mL PCR tubes. Positive control with known template and appropriate, size was

Table 2

Susceptibility *B. fragilis* group species to eleven antimicrobial agents.

Antibiotics ^a	Distribution of <i>B. fragilis</i> group isolates according to break point ^a %		
	Susceptibility N/%	Intermittent N/%	Resistance N/%
Ampicillin/Sulbactam	54(87%)	3(4.8%)	5(8%)
Cefoxitin	27(43.5%)	2(3.2%)	33(53.2%)
Ceftriaxone	25(40.3%)	1(1.6%)	36(58%)
Chloramphenicol	56(90.3%)	1(1.6%)	5(8%)
Clindamycin	25(40.3%)	1(1.6%)	36(58%)
Ertapenem	52(84%)	3(4.8%)	7 (11.5%)
Imipenem	31(50%)	2(3.2%)	29(46.7%)
Meropenem	49(79%)	2(3.2%)	11(17.7%)
Metronidazole	55(88.7%)	5(8%)	2(3.5%)
Moxifloxacin	32(50.8%)	1(1.6%)	29(46.7%)
Piperacillin/Tazobactam	62(100%)	-	-

^a Breakpoint used in accordance with CLSI M11 9th Edition, 2018.

prepared. The thermocycler (Agilent Technologies, Sure cycler 8800) was programmed according to the standardized set up (Table 1).

The amplification products were analysed in 1.5% agarose gel in 1x TAE buffer with ethidium bromide. The electrophoresis power was supplied at 90–100 voltage for 1 h and images of the agarose gel was captured using UVP software.

3. Results

Results of comparative in vitro activity of antimicrobial agents against the randomly selected sixty-two pure culture *B. fragilis* group strains are summarised in Table 2. During the period of this study, phenotypically, the antibiotics showed activity against the isolates but noticeable increase in resistance was observed. Among the β-lactam agents tested cefoxitin and ceftriaxone exhibited high resistance in all the species while β-lactam inhibitor has shown more activity. Carbapenems resistance is also observed in all species, higher resistance rate is seen in imipenem with MIC of ≥16 µg/ml. Overall, clindamycin has exhibited the least activity and a rapid climb in resistance with MIC of ≥12 µg/ml. Resistance to moxifloxacin was notably high. Metronidazole has shown two resistance strains with MIC value of >32 µg/ml; *B. fragilis* and *B. caccae*, moreover, intermittent resistance was exhibited in 5 strains (12–16 µg/ml); three *B. fragilis* and two *B. distansoni*.

Molecular detection of the four target resistance genes (*nim*, *cfiA*, *ermF* and *cepA*) were expressed at various degree among the tested 62 *B. fragilis* group strains. The detailed results are shown in Table 3 (See Fig 1–4). Some strains expressed a combination of two or more resistance genes such as two genes 25.8%, three genes 33.8% and four genes 3.2%. *B. fragilis* was found to be more resistant compared to other members of the group. Among the isolates harbouring 2 genes, the more common combination was *ermF* and *cepA* while *cfiA*, *ermF* and *cepA* were co-existing in

Table 1

List of DNA Primers sequence used for this study (Bioserve, Hyderabad).

Gene Target ^a	Sequence (5'–3')	Amplicon size (bp)	Reference	PCR cycles (35 cycles) ^b
<i>Nim</i>	5'-ATGTTTCAGAGAAATGCGCGTAAGCG-3' 5'-GCTTCCTTGCCTGTGATGTGCTC-3'	458	47	94 °C 30s, 57 °C 30 s, 72 °C 45s
<i>cepA</i>	5'- TTT CTG CTA TGT CCT GCC C-3' 5'- ATC TTT CAC GAA GAC GGC-3'	780	11	94 °C 30s, 60 °C 30 s, 72 °C 45s
<i>cfiA</i>	5'-ATGGTACCTTCCAACGGG-3' 5'-CACGATATTGTCGGTGC-3'	353	11	94 °C 30s, 57 °C 30 s, 72 °C 45s
<i>ermF</i>	5'-CGGGTCAGCACTTTACTATTG-3' 5'-GGACCTACCTCATAGACAAG-3'	466	13	94 °C 30s, 50 °C 30 s, 72 °C 45s

^a *Nim*-metronidazole, *cfiA*-cephalosporins, *ermF* -clindamycin, *cepA* -carbapenem

^b PCR cycle was standardized at Yenepoya Research Centre, Mangalore, India.

Table 3
Distribution of target resistance genes in *B fragilis* group (62 strains).

Organism/Species	Number of species	Target gene of antibiotic resistance			
		nim	cfiA	ermF	cepAa
<i>B fragilis</i>	36	7(11.2%)	20(32.2%)	23 (37%)	16 (25.8%)
<i>B thetaiotamicron</i>	11	1(1.6%)	7(11.2%)	8 (12.9%)	7 (11.2%)
<i>B ovatus</i>	3	0	1(1.6%)	1(1.6%)	1(1.6%)
<i>B distasonis</i>	7	2(3.2%)	6(9.6%)	4(6.4%)	3(4.8%)
<i>B caccae</i>	5	2(3.2%)	2(3.2%)	3(4.8%)	3(4.8%)
Total	62	12(19.3%)	36(58%)	39 (62.9%)	30(48.3%)

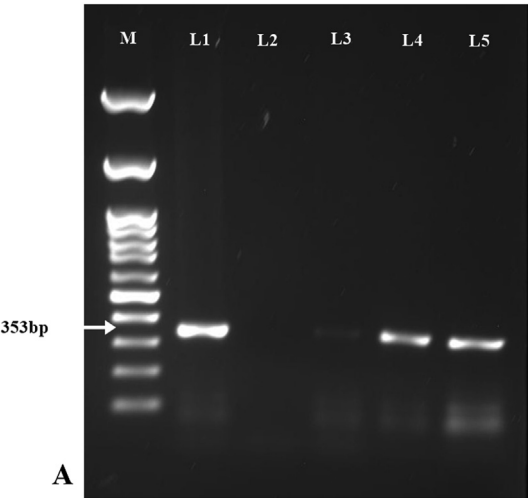


Fig. 1. Amplification of *B fragilis* group specific *cfiA* gene (353bp) M-100bp ladder, Lane 1,3,4 & 5 showing resistant gene, Lane 2 absence of genes.

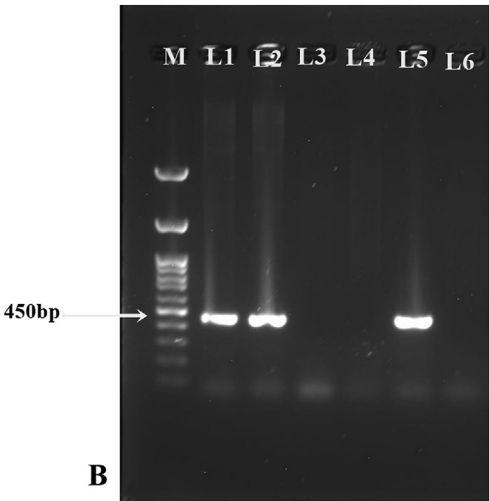


Fig. 2. Amplification of *B fragilis* group specific *nim* gene (450bp) M- 100bp ladder, Lane 1 positive known *nim* gene, Lane 2&5 shows presence of *nim* gene, Lane 3,4 & 6 absence of *nim* gene.

the isolates with 3 genes.

4. Discussion

Although the emerging antimicrobial resistance of *B fragilis* group have been reported from western countries, only few data is available from India [20–25].The leading concern of antibiotic resistance is due to misuse of antibiotics causing insufficient and failure of treatment with

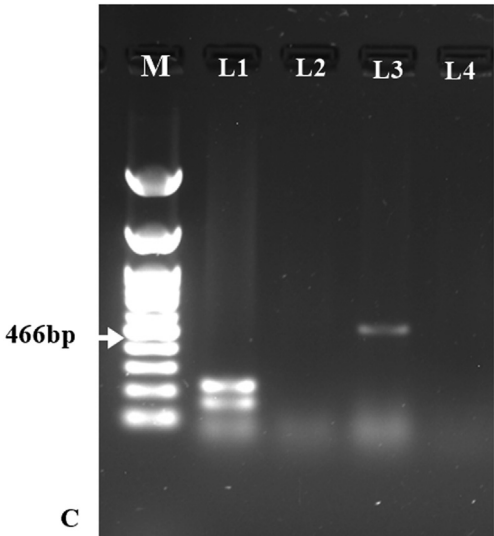


Fig. 3. Amplification of *B fragilis* group specific *ermF* gene (466bp) M-100bp, Lane 3, showing presence of *ermF* gene, Lane 1,2&4 absence of gene.

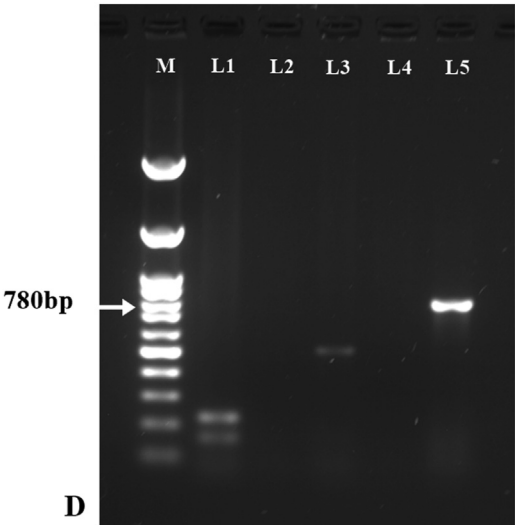


Fig. 4. Amplification of *B fragilis* group specific *cepA* gene (780bp) M-100bp, Lane 5 showing presence of *cepA* gene, Lane 1,2, 3, 4 absence of gene.

mortality rate of 45% among these cases [26].Resistance pattern differs based on possible factors in association with species, specimen, antibiotic consumption, hospitals and geographical region [3,27].According to Boyanova et al., antibiotic evolution of resistance to *B fragilis* group was observed and highly increased in β -lactam agents and clindamycin while it was rare in carbapenems and no evolution was seen in metronidazole [3].

In one of our previous study, β lactamase producers were detected in 77.7% out of 72 strains using cefinase disc which is in accordance with the present findings. [28] Brook et al. has also reported that due to the indigenous cephalosporinase production, cephalosporin has insubstantial use against members of *B. fragilis* group [4]. The enzyme is mainly encoded by *cepA* gene which is present in 90% of the isolates [29]. The carbapenemase *cfiA* confers resistance to practically all β -lactams [30, 31]. Although carbapenem resistance was reported as rare [3], it was also reported that Asian countries have higher resistance to carbapenems (>10%) than other western countries (>5%) [32].

The various levels of carbapenem resistance is due to the different range of expression of the *cfiA* genes. Other studies have given the impression that *B. fragilis* strains that possess *cfiA* gene appears to be susceptible to carbapenems but with high potential to get converted into high-level β -lactam resistance [31]. Higher levels of resistance are seen preferentially in *B. fragilis* when the *cfiA* gene expresses efficiently by acquisition of four specific (IS1186, IS942, IS1224 and IS4351) insertion sequence upstream in the promotor region of the gene which is expressed specifically for carbapenem resistance as suggested by Gutacker et al. [11]. In the present study, out of 62 strains, *cfiA* and *cepA* genes were present in 58% and 48.3% respectively, which is in accordance with Nakano et al. [33].

In the course of the years, the over use of clindamycin has resulted in a drastic increase in resistance and has been deleted from recommended drugs to treat severe anaerobic infections empirically, as documented in the literature [34]. Studies have shown high prevalence of clindamycin resistance which correlates mainly with the *ermF* gene and shows concordant results with the present finding [35–37]. Eitel et al. reported that regardless of sensitivity of clindamycin, the occurrence of the *ermF* gene in *B. fragilis* and non - *fragilis* strains were indistinguishable [36]. Other investigators had also reported the possibility of higher degree of *ermF* gene due to the association with conjugative transposon. It is also noted that the host range extends as other bacteria carrying one or more *erm* genes other than *ermF* gene act as a reservoir in both human, animals and related conjugative elements found outside *B. fragilis* group. [13].

The first clinical isolate of *B. fragilis* resistant to metronidazole, the drug of choice to anaerobes was reported in 1978 due to long term therapy [37], followed by other investigators from different countries [38–47]. Metronidazole resistance was attributed to the association of *nim* gene in these studies. However, Thrinh & Reyssset et al. in their findings revealed that four related *nim* (A-D) genes are located at mobilizable plasmid which was transferable by conjugation. This process was presumed to occurred in vivo and have been detected in different clinical *B. fragilis* isolates [48]. Although the encoding *nim* gene is the cause of resistance, Nagy et al. has shown that Enterococci may cause the inactivation of metronidazole during co-cultivation with *B. fragilis* which will shelter *B. fragilis* in mixed cultures [49], which might result in therapeutic failure. In the current study, 12 out of 62 strains were positive for *nim* gene, among this only one strain inhibited metronidazole at an MIC value of >32 μ g/ml, while other strains were sensitive. The strains that inhibited metronidazole intermediately did not express *nim* gene. This finding indicates that other determinant factors must have played a role in expressing the metronidazole resistance.

The first Indian report on metronidazole resistance was authored by Chaudhary et al., in 2011 [20]. Following this few more case reports [23] sequencing studies [22, 24] molecular detection of *nim* gene by PCR [21, 25] had emerged. However, the data regarding the co-existence of MDR genes is lacking from India. The results of the present study might throw some light into the unexplored area.

5. Conclusion

This study explores the antibiogram and occurrence of resistance genes in *B. fragilis* group from various clinical samples and found a decent amount of antibiotic resistance. Speciation level of identification in

Bacteroides strains can also play an important role clinically as antibiotic resistance is expressed at different rate by different species. We can also conclude that *B. fragilis* strains harbouring the resistant gene may not fully be expressed in vitro tests. More investigation is required to overcome these uncertainties. Thus, antimicrobial profiling and regular monitoring of anaerobes should be carried out more often to avoid future resistance.

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None.

Declaration of competing interest

There is no conflict of interest to declare.

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